

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

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Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed transfer pairs among Chapman–Shaoxing, CPSC2018+2019, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (BCa, $B = 10,000$) on the full 60-experiment grid. The study followed a pre-registered protocol with dated, publicly archived amendments. The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60 (75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met (23/60 = 38.3%), triggering the registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (+0.0159 vs.

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+0.0083). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$); and a discrimination-aware gate (C5) shows 7/12 pair $\times$ architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [?, ?, ?]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

For ECG this question is not academic. PTB-XL [?], Chapman-Shaoxing [?], and CPSC use different acquisition protocols, lead sets, and label vocabularies, so a model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [?]. Clinicians and regulators then face a concrete decision: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [?]. Fitted under a convex loss, it minimizes NLL on the calibration split, so its calibration benefit there is empirically small; on a shifted target the source-fit temperature is mismatched to the target logits and a non-trivial

benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (nine negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. ??). To avoid HARKing, protocol v2.1-A1 [?] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from decay to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [?, ?]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported.

Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, triggering the “ID nonzero boundary” branch (protocol A1.4.2): the premise that TS has no ID recalibratable miscalibration is re-examined, the

73 decay contrast restored as exploratory, and C1 qualified, since the OOD
 74 benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific.
 75 $47/60 = 78.3\%$ of ID point estimates are positive and $2/60 = 3.3\%$
 76 have $\Delta\text{ECE}_{\text{ID}} < -0.01$ (counter-example 6, -0.0152 ; one marginal case,
 77 -0.0128).

78 **C3 (Methodological, Shapley decomposition validation):** A 27-cell full-
 79 factorial synthetic validation (slope $\times$ intercept $\times$ prevalence, $n = 20,000$)
 80 recovers ground truth in the identifiable regime (27/27 at $n = 20,000$;
 81 21/27 at $n = 500$), flags entangled regions without a threshold, and reports
 82 bootstrap CIs on absolute Shapley values.

83 **C4 (Practical, method-selection boundary):** Across 8 methods $\times$ 6 pairs $\times$ 5
 84 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior adaptation is
 85 fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included (InceptionTime
 86 15/30); after exclusion the EM OOD finding attenuates. The mechanism
 87 is prior-likelihood mismatch: EM fits a target prior that amplifies shift-
 88 induced miscalibration when the source confusion matrix is ill-conditioned.

89 **C5 (Clinical, discrimination-aware calibration gate):** A two-layer gate decou-
 90 ples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viabil-
 91 ity (accuracy $\geq$ source-domain baseline). Across 12 pair $\times$ architecture
 92 strata, 7/12 pass both layers; 5/12 fail the discrimination layer, including
 93 directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below baseline (e.g., PTB-
 94 XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$, $\text{acc} = 0.657 < 0.713$). The
 95 gate reframes the 9/60 counter-examples: 4/9 calibration failures (C1 re-
 96 jects) and 5/9 discrimination failures (C5 rejects despite C1 passing).

97 Prior ECG calibration studies report the phenomenon [?, ?, ?, ?] but not an
 98 actionable *when does TS pay off* answer with multi-seed robustness evidence.
 99 The novelty is *not* multi-seed validation on ECG (a methodological choice) but
 100 the **first systematic characterization of the boundary conditions under which**
 101 **TS yields a positive OOD calibration benefit in cross-corpus ECG transfer:**
 102 the pre-registered OOD quantification, the ID boundary condition, the method-
 103 selection boundary (TS robust; EM prior and Matrix scaling weakest), the pre-
 104 dictability failure, and the discrimination-aware gate. That characterization is the
 105 actionable contribution; multi-seed validation is the evidence standard.

106 2. Related Work

107 2.1. Calibration Estimation and Recalibration

108 The calibration of modern neural networks has been studied extensively since
 109 the seminal work of Guo et al. [?], who showed that deep networks are often
 110 overconfident and introduced temperature scaling (TS) as a simple post-hoc recal-
 111 ibration method. Nixon et al. [?] proposed the static calibration error, and Kumar
 112 et al. [?] introduced verified uncertainty calibration with distribution-free guaran-
 113 tees. Kull et al. [?] generalized TS to Dirichlet calibration for multi-class settings.
 114 In the clinical risk-model context, Van Calster et al. [?, ?] established a calibration
 115 hierarchy from utopia to empirical data and argued that calibration is the “Achilles
 116 heel” of predictive analytics. Strodtzoff et al. [?] benchmarked deep learning for
 117 ECG analysis on PTB-XL and noted calibration degradation under subpopulation
 118 shift, motivating our cross-corpus design.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

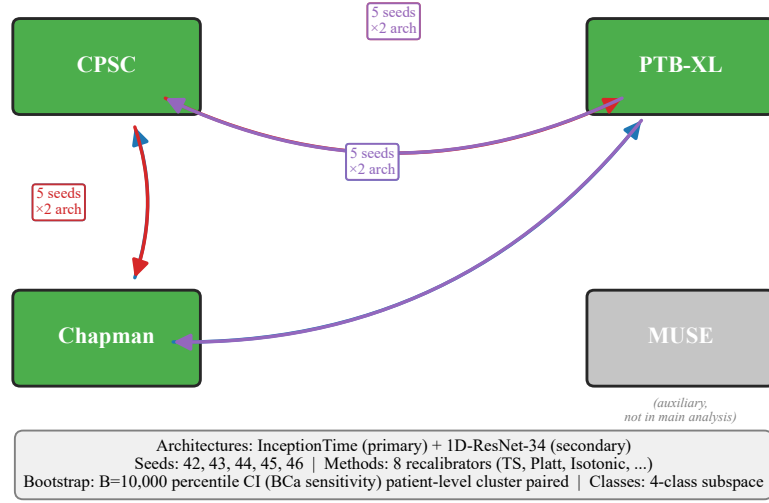


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, ResNet1D) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

119 2.2. *Prior-Shift Estimation and Transfer Calibration*

120 Lipton et al. [?] introduced black-box shift estimation (BBSE) for detecting
121 and correcting label shift using confusion-matrix inverses. Alexandari et al. [?]
122 showed that maximum-likelihood calibration with bias correction is hard-to-beat
123 at label shift adaptation, providing the theoretical foundation for our EM prior-
124 shift method. For transfer calibration upper bounds, TransCal [?] estimates trans-
125 ferability for auto-selecting target-domain unsupervised optimal T ; PseudoCal [?]
126 uses pseudo-labels for calibrated transfer; and LaSCal [?] provides label-aware
127 shift calibration with distribution-free guarantees. Our S1 paradigm differs fun-
128 damentally: we fit on the *source* calibration split only (no target labels), with
129 S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly
130 weaker assumption than TransCal/PseudoCal, which access target logits or labels.

131 2.3. *Cross-Domain ECG Calibration*

132 Barandas et al. [?] evaluated uncertainty quantification methods in multi-
133 label ECG classification, finding that ensemble-based methods outperform single-
134 model approaches under subpopulation shift. Vranken et al. [?] studied uncer-
135 tainty estimation for 12-lead ECG deep learning and reported significant calibra-
136 tion drift across acquisition devices. Haekal et al. [?] performed cross-dataset
137 validation for atrial fibrillation detection with calibration-aware probability analy-
138 sis, and Patel & Beedala [?] documented calibration drift under cross-institutional
139 ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general
140 vision [?]; we contribute the multi-seed ECG quantification and the boundary-
141 condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

142 2.4. ECG Foundation Models and Transfer Learning (2024–2026)

143 Recent work has introduced ECG foundation models trained on millions of
144 recordings. ECGFounder [?, ?] (10 million ECGs, Harvard–Emory) reaches
145 expert-level performance on 80 diagnoses and transfers cross-domain via fine-
146 tuning. ECG-FM [?] is an open-weight transformer pretrained on 1.4 million ECG
147 segments with hybrid contrastive and generative self-supervision, showing strong
148 cross-dataset generalizability and label efficiency. HeartLang [?] treats heartbeats
149 as words and rhythms as sentences via a QRS-Tokenizer and masked sentence pre-
150 training. Tan et al. [?] found that fine-tuning benefits diminish as the downstream
151 dataset grows and that CNNs transfer more effectively than RNNs in multi-label
152 ECG classification. For OOD adaptation, ADAPTOOD [?] quantifies distribution-
153 shift severity via uncertainty estimation to guide selective LoRA layer unfreez-
154 ing, and BeatRhythm-TTA [?] performs test-time adaptation with SQI-gated self-
155 training and beat-rhythm consistency. Costa et al. [?] introduced STAR (Sinu-
156 soidal Time–Amplitude Resampling), a morphology-preserving beat-wise aug-
157 mentation for cross-source generalization, and Chen et al. [?] proposed PEACE
158 for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment.
159 These works advance ECG transfer but not the *calibration* of transferred models,
160 the gap our study fills via pre-registered cross-corpus evaluation of recalibration
161 methods.

162 3. Data and Pre-registered Protocol

163 Figure ?? summarizes the experimental design.

164 3.1. Datasets

- 165 • **PTB-XL v1.0.3** [?]: the official release contains 21,837 records; the down-
166 loaded `ptbx1_database.csv` copy contains 21,799 (a 38-record deficit in
167 the source CSV, disclosed as a provenance limitation), official 10-fold (folds
168 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records
169 are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- 170 • **Chapman-Shaoxing** [?, ?]: 20,265 raw records; 1 signal-read error
171 removed and 21 non-finite records quarantined (`qc_report.json`), for
172 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- 173 • **CPSC2018+2019**: patient-wise split; HYP class extremely rare ($n = 11$) →
174 main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides
175 of the pair recomputed symmetrically.

176 3.2. Label Mapping and Leakage Guard

177 Single-label priority rules are fixed (MI > STTC > CD > HYP
178 > NORM) and sensitivity-analyzed [?, ?]. Sinus rate/form statements
179 (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because
180 PTB-XL's `scp_statements.csv` lists them as rhythm statements with no
181 `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard
182 assertion in the reproducibility repo). Normalization statistics come only from
183 the train split; augmentation applies only to train.

184 4. Methods

185 4.1. Transfer pairs and shift matrix

186 We study six directed cross-corpus transfer pairs: Chapman→CPSC,
187 Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman,
188 and PTB-XL→CPSC. The class count is *not* uniform across pairs and is fixed per
189 pair before any endpoint is computed, as follows. A pair whose source or target is
190 CPSC2018+2019 is restricted to the 4-class subspace {NORM, CD, STTC, MI}
191 (CPSC’s HYP class has $n = 11$, §??), with both sides of the pair recomputed sym-
192 metrically: this covers Chapman→CPSC, CPSC→Chapman, CPSC→PTB-XL,
193 and PTB-XL→CPSC ($K = 4$). The two pairs connecting Chapman–Shaoxing
194 and PTB-XL, which contain no CPSC side, retain the full 5-class vocabulary
195 {NORM, MI, STTC, CD, HYP}: Chapman→PTB-XL and PTB-XL→Chapman
196 ($K = 5$). Each experiment records its own K and label mapping (num_classes,
197 label_map in transfer_result.json), so no endpoint mixes class counts
198 within a cell. Consequently the primary endpoint is a top-label ECE on a 4- or
199 5-class task depending on the pair, and the pooled $\Delta\text{ECE}_{\text{OOD}}$ mixes both K values
200 across cells; the per-pair tables (Tables ??–??) are therefore the appropriate level
201 at which to read K -specific results. The L2 eval-time shift matrix (no retraining)
202 covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise
203 injection (BWL/MA/EM) at SNR {24, 12, 6, 0, -6} dB; gain $\times 0.5 / \times 2$.

204 4.2. Architecture and seeds

205 We use two architectures: InceptionTime (primary, 46,000–184,000 param-
206 eters) and ResNet1D (a 1D-ResNet-34 backbone) (secondary, with full main-
207 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The Bi-

208 Mamba architecture is included only as a toy experiment ($n = 60$ records, down-
 209 sampled resolution) and deferred to the supplement; the full-resolution BiMamba
 210 run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the
 211 main-endpoint analysis or the architecture-robustness claim. Training: AdamW +
 212 cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training;
 213 temperature applied only as post-hoc temperature scaling.

214 4.3. Calibration methods and scenarios

215 We compare the no-recalibration baseline (None) against eight recalibration
 216 methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scal-
 217 ing, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and
 218 BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline
 219 plus these eight recalibration methods constitute the delivered set; CORAL, TTA,
 220 MC-Dropout, and Oracle are not delivered on real data (a registered deviation,
 221 see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply),
 222 the main analysis scenario (Sec. ??); **S1'** denotes the target-domain unsupervised
 223 variant in which EM/BBSE estimate the target prior from unlabeled target predic-
 224 tions.

225 4.4. Pre-registered endpoints

226 Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [?]):

$$\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}} \quad (1)$$

227 where $ECE_{\text{raw}}^{\text{OOD}}$ is the 4-class-task top-label confidence SmoothECE on the tar-
 228 get corpus with raw softmax probabilities, and $ECE_{\text{TS}}^{\text{OOD}}$ is the same after source-
 229 cal-fit temperature scaling. Following the standard convention of Guo et al.

[?], ECE (and its SmoothECE variant) is evaluated on the *top-label* marginal: the predicted maximum probability against 0/1 top-label correctness; classwise-ECE is reported separately as a secondary endpoint. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta \text{ECE}_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta \text{ECE}_{\text{OOD}} > 0 \quad (2)$$

The one-sided alternative $H_1 : \Delta \text{ECE}_{\text{OOD}} > 0$ follows the empirical asymmetry registered at revision A1: on ID-like data the source-fit temperature yields, empirically, little calibration change, whereas on a shifted target it is mismatched and a non-trivial benefit is often retained [?, ?]. This direction prior is empirical, *not* a theorem: TS applied to an under-confident target can increase ECE, and our own data contain nine negative cells. The choice was made before the pilot results, which only triggered the endpoint re-location (revision A1). The one-sided test is the pre-registered choice, archived before multi-seed unblinding. Patient-level cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [?]; stratified pooling: strata = transfer pair $\times$ architecture.

The pre-registered CI method is BCa [?], at $B = 10,000$, re-run on the entire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The percentile bootstrap is the large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$; for the paired-cluster design with $B = 10,000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were run ($B = 10,000$ percentile re-runs), BCa and percentile CIs agree in sign and magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain as historical provenance (5 lack log provenance, see Limitations); percentile CIs serve only as a sensitivity cross-check.

254 Boundary-condition endpoint:

$$\Delta\text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

255 The pre-registered expectation is that $H_0 : \Delta\text{ECE}_{\text{ID}} = 0$ is *not rejected*; the criterion
256 is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid*
257 *re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered*
258 *(percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. ??).*

259 Secondary endpoints: G (recoverable loss vs. oracle; *not delivered* — the Or-
260 acle recalibrator was implemented (fit_oracle/apply_oracle) but never ex-
261 ecuted on real data, so G is undefined here and is excluded from all reported
262 results; see Limitations), decay = $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descrip-
263 tive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept,
264 predicted/true prior L_1 .

265 Family structure: BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$ 13 shift levels =
266 156 primary comparisons; the multi-seed validation is robustness evidence (60
267 seed experiments), not a confirmatory family.

268 4.5. Pre-registration revision A1 (transparency statement)

269 The primary endpoint was re-located from decay to $\Delta\text{ECE}_{\text{OOD}}$ via pre-
270 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent
271 exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the
272 original ID-vs-OOD contrast. The revision direction is consistent with the empir-
273 ically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived
274 as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes.
275 The pilot results do not enter the main test report. All hypotheses in this paper are
276 tagged *exploratory + robustness validation*, not confirmatory.

277 The original BH-FDR family of 156 comparisons ($6 \text{ pairs} \times 2 \text{ architectures} \times$
 278 13 shift levels) is reduced to a confirmatory family of 12 hypotheses ($6 \text{ pairs} \times 2$
 279 $\text{architectures} \times 1 \text{ primary shift level}$) under amendment A2, with the remaining
 280 13 shift levels designated as exploratory dose-response. Under the reduced family,
 281 BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12
 282 yields 2/12. The A2 amendment is archived in docs/PROTOCOL_AMENDMENT_A2_
 283 FAMILY_REDUCTION.md.

284 4.6. *Decomposition of the benefit decay*

285 We decompose the retained benefit into slope / intercept / prevalence contribu-
 286 tions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

287 where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the
 288 chain residual (zero under nested counterfactuals). Identifiability is checked via
 289 the Fisher information determinant on a pre-registered threshold τ ; entangled re-
 290 gions are reported on the error map without a threshold. Shapley attribution is
 291 reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$)
 292 mark uninterpretable shares.

293 4.7. *Two-layer joint bootstrap*

294 Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$
 295 joint ΔECE CI (two-layer; honest non-nested approximation), per protocol §7.

296 4.8. *Net Clinical Value (NCV)*

297 Net Clinical Value (NCV), a pre-registered secondary endpoint, measures
 298 the clinical decision impact of TS. For each (sample, class) pair we compare

the binary decision (threshold $\tau = 0.5$) before and after calibration, counting TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves clinical decisions. It is computed on ID and OOD splits (ncv_id, ncv_ood; E3 CSV).

4.9. Tooling

Parts of the analysis and plotting code were drafted and refactored with the assistance of a widely available large language model service, used as a supervised coding assistant. All experiments and statistical analyses were executed with the author’s own scripts, and every reported number is reproducible from the public repository (Code availability). The author reviewed the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure ?? visualizes the shift-dose response; Table ?? reports the primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{\text{OOD}} = 0$ vs $H_1 : \Delta ECE_{\text{OOD}} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

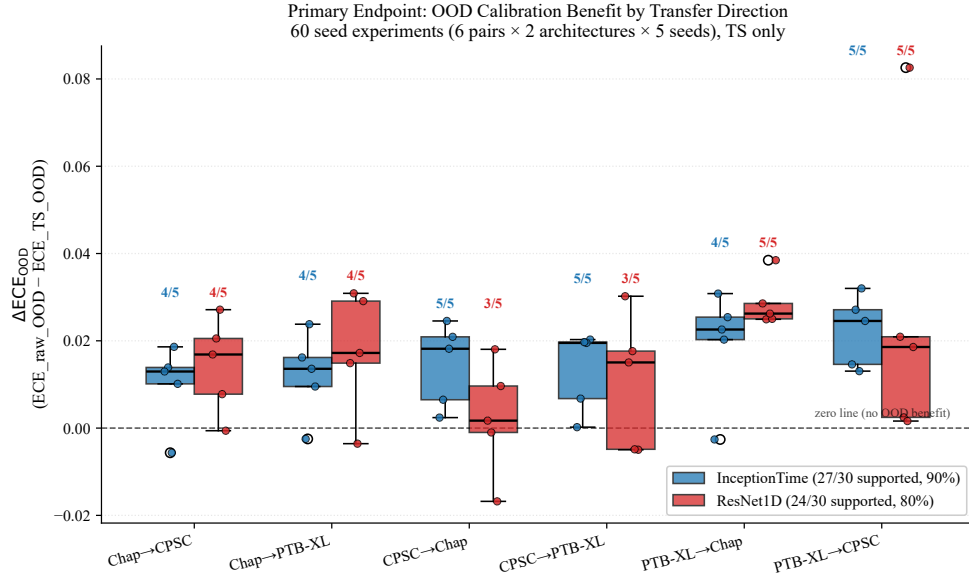


Figure 2: Primary endpoint ΔECE_{ODD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. Class count $K = 4$ for the four pairs involving CPSC (Chap $\rightarrow$ CPSC, CPSC $\rightarrow$ Chap, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC) and $K = 5$ for the two CPSC-free pairs (Chap $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chap); see §?? . “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap $\rightarrow$ CPSC ($K=4$)	IT	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	RN	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap $\rightarrow$ PTB-XL ($K=5$)	IT	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
CPSC→Chap ($K=4$)	RN	42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
	IT	42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
CPSC→PTB-XL ($K=4$)	RN	42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
	IT	42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓
		44	+0.0158	+0.0195	+0.0189	✓
		45	+0.0052	+0.0197	+0.0193	✓
		46	+0.0002	+0.0002	+0.0002	✓
	RN	42	+0.0114	+0.0150	+0.0148	✓
		43	+0.0048	−0.0050	−0.0051	✗
		44	−0.0032	−0.0049	−0.0049	✗
	IT	45	+0.0130	+0.0176	+0.0173	✓
		46	+0.0261	+0.0302	+0.0296	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
PTB-XL→Chap ($K=5$)	IT	42	+0.0239	+0.0308	+0.0275	✓
		43	+0.0245	+0.0254	+0.0225	✓
		44	+0.0173	+0.0203	+0.0195	✓
		45	+0.0162	+0.0226	+0.0200	✓
		46	−0.0011	−0.0026	−0.0029	✗
	RN	42	+0.0145	+0.0249	+0.0191	✓
		43	+0.0338	+0.0385	+0.0336	✓
		44	+0.0213	+0.0250	+0.0228	✓
		45	+0.0228	+0.0262	+0.0231	✓
		46	+0.0239	+0.0286	+0.0221	✓
PTB-XL→CPSC ($K=4$)	IT	42	+0.0196	+0.0271	+0.0244	✓
		43	+0.0166	+0.0245	+0.0222	✓
		44	+0.0183	+0.0320	+0.0303	✓
		45	+0.0006	+0.0131	+0.0105	✓
		46	+0.0097	+0.0146	+0.0123	✓
	RN	42	+0.0010	+0.0025	+0.0023	✓
		43	+0.0197	+0.0186	+0.0145	✓
		44	+0.0635	+0.0826	+0.0772	✓
		45	+0.0001	+0.0016	+0.0015	✓
		46	+0.0126	+0.0209	+0.0190	✓

320 Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%,
321 ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence (10/10,
322 zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed 45). The
323 nine counter-examples are explained in Sec. ??.

324 The 60 experiments are balanced (6 directions $\times$ 2 architectures $\times$ 5 seeds, no
325 partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture interaction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5 on both architectures (10/10).

5.2. ID boundary condition: ΔECE_{ID}

Table ?? reports the boundary-condition endpoint ΔECE_{ID} for TS; the pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID} has mean +0.0083 (vs. OOD mean +0.0159, a $1.9\times$ ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast is restored as an exploratory endpoint; and C1 is

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

349 qualified, the OOD benefit being about $1.9\times$ the ID benefit rather than an exclu-
350 sively OOD effect. The positive ID estimates reflect finite-sample calibration-split
351 variation, SmoothECE estimator bias, and a val-ECE early-stopping effect (model
352 selection on the endpoint family), all disclosed.

353 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

354 Table ?? reports the 8-method $\times$ 6-pair support rate over 5 seeds on Inception-
355 Time. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE achieves 22/30

356 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most
 357 fragile on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood
 358 mismatch mechanism (C4). Across 2 architectures (Table ??), TS achieves 51/60
 359 (85.0%) (InceptionTime 27/30, 90.0%; ResNet1D 24/30, 80.0%), with full 5-seed
 360 coverage on all 6 directions (Figure ??).

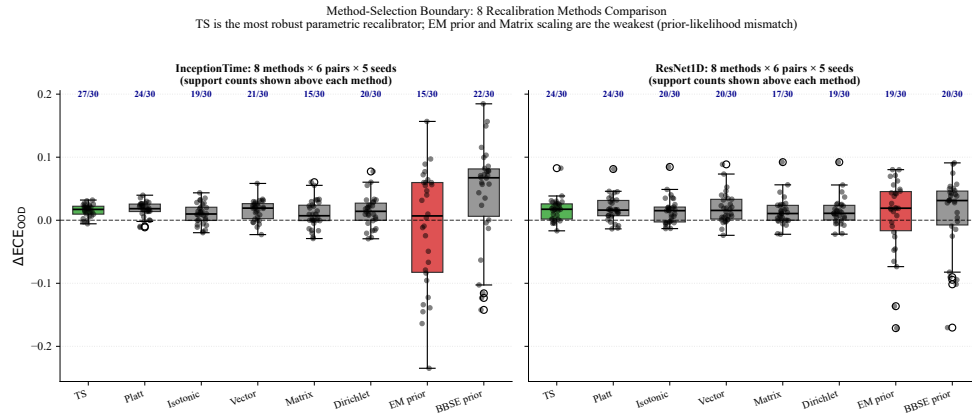


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs
 $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior
 and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30,
 Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g.,
 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

361 Across the 30 TS experiments per architecture, the standard deviation of
 362 ΔECE_{OOD} is 0.0098 (InceptionTime) and 0.0180 (ResNet1D), a $1.84\times$ ratio, with
 363 ResNet1D’s highest per-direction variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, in-
 364 cluding an unexcluded favorable outlier at seed 44, $\Delta ECE_{OOD} = +0.0826$). Sup-
 365 port deficits concentrate in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a
 366 direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields

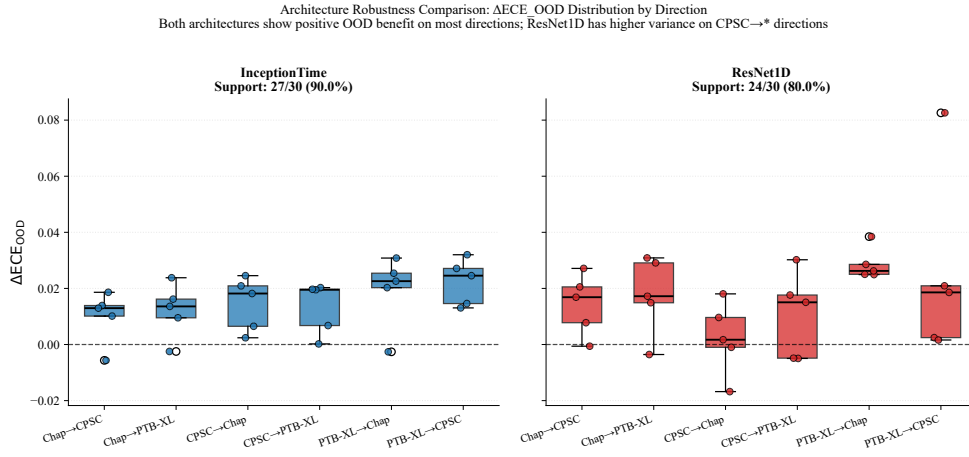


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (InceptionTime). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat*: 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

367 $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%)
368 is not significant at $\alpha = 0.05$.

369 EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal,
370 then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated
371 confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies
372 rather than corrects the shift-induced miscalibration. EM reaches only 15/30 sup-
373 port on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-
374 signature-specific), not method-intrinsic.

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

375 *Why TS and not Platt..* Platt achieves a marginally higher mean OOD benefit
376 than TS (+0.0176 vs. +0.0159; InceptionTime +0.0169 vs. +0.0152, Table ??),
377 which invites the question of why TS is singled out as the recommended default.
378 Three measured criteria separate them, and all three favour TS. First, *robustness*
379 *of the sign*: TS supports 51/60 cells versus Platt’s 48/60, i.e. TS fails to certify
380 a positive benefit in three fewer experiments, which is the quantity a deployment
381 decision actually depends on. Second, *dispersion*: TS has a smaller across-cell
382 standard deviation (0.0146 vs. 0.0167, a 13% reduction), and the methods with
383 the highest mean benefit are markedly less stable (BBSE +0.0266 at sd 0.0736,
384 EM −0.0026 at sd 0.0793) — a high mean bought with a heavy tail is the worst

385 profile for a per-deployment recalibrator. Third, *the difference is not statistically*
 386 *distinguishable*: the patient-level paired contrast over the 60 matched cells gives
 387 $\overline{\Delta ECE}_{\text{TS}} - \overline{\Delta ECE}_{\text{Platt}} = -0.0017$ (sd 0.0117, paired $t = -1.14$, $p \approx 0.26$), with
 388 TS the larger benefit in exactly 30/60 cells — a coin flip. The mean advantage of
 389 Platt is therefore inside the noise, whereas the support advantage of TS is not a
 390 noise-level quantity: it is a count of cells in which the sign is certified. Given that
 391 the two methods are statistically indistinguishable on magnitude, the tie is broken
 392 on certification rate and dispersion, both of which favour TS. A further structural
 393 point reinforces the choice: TS is a single-parameter monotone transform and
 394 therefore exactly argmax-preserving, whereas per-class Platt and isotonic are fit-
 395 ted independently per class, which generates each class’s score from a different
 396 monotone map and so does *not* preserve the ranking within a sample (the imple-
 397 mentation declares this explicitly in its docstring). TS is thus the only method
 398 in the delivered set whose calibration guarantee is obtained without perturbing
 399 the discrimination it leaves untouched (§??). None of this makes Platt wrong; it
 400 makes Platt a reasonable alternative whose mean advantage is not resolvable at
 401 $n = 60$, and whose certification rate is lower.

402 5.4. Shapley decomposition validation (27 cells, $n = 20,000$)

403 The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in$
 404 $\{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposi-
 405 tion recovers ground truth within the pre-registered identifiable regime (Fisher
 406 information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the
 407 recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of
 408 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in ex-
 409 treme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most

410 clinically significant (see `decomposition_validation.csv`; the $n = 20,000$
 411 CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three
 412 entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\det \in \{2.4 \times$
 413 $10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with
 414 the recovery-failed flag set. Bootstrap CIs on absolute Shapley values and share-
 415 reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

416 The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE;
 417 its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of
 418 decomposition explanatory power) remains to be run on real data; per the protocol
 419 self-binding clause, “mechanism” wording is limited to “mechanism decomposi-
 420 tion (synthetic validation part)” until cross-fitting completes.

421 5.5. Predictability experiment (*failure branch triggered*)

422 The predictability experiment (protocol §8.5) tests whether an unsupervised
 423 target-domain statistic (predicted-prior L_1 distance, logit moments) predicts ben-
 424 efit retention via leave-one-fold-out R^2 with bootstrap CI (Table ??). The imple-
 425 mented fold unit is (architecture, transfer pair) (12 folds), versus the registered
 426 pair-only folds (6); under that registered deviation the pair-only recomputation re-
 427 tains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime +
 428 ResNet1D); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

429 Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$.
 430 The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5 , trig-
 431 gering the **failure branch**: the three-step chain (decomposition + predictability +
 432 deployment) is downgraded to a **two-step chain** (decomposition + deployment),
 433 and the deployment criterion is renamed “empirical lookup table” rather than “pre-
 434 dictive criterion”. No post-hoc predictability claim is made.

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

Re-running the LOO R^2 with three label-free features (fitted temperature T , Wasserstein-1 distance between source and target logit distributions, class-wise entropy difference), all fail against $\Delta\text{ECE}_{\text{OOD}}$: T yields $R^2 = -0.13$, Wasserstein $R^2 = -0.06$, entropy $R^2 = -0.02$, all three combined $R^2 = -0.11$. The predictability boundary is *robust*: no label-free feature, including T , predicts $\Delta\text{ECE}_{\text{OOD}}$ above $R^2 = 0.5$. This strengthens the failure branch, and the gate in §??, based on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the appropriate actionable criterion.

5.6. Deployment criterion (sensitivity / specificity / DCA / safety)

We evaluate the deployment criterion (protocol §9) on the L2 shift matrix: predict “deploy TS” when raw ECE exceeds a threshold, and report sensitivity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS

447 SmoothECE ≤ 0.05 and Δ ECE CI lower bound > 0 , primary-endpoint conven-
448 tion). Comparators: always-none, always-TS, always-buy- n -labels.

449 Sensitivity / specificity (Table ??): Youden’s J is maximized at raw ECE
450 $= 0.127$, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sensitivity is
451 high (92% of beneficial cases caught) but specificity low (20% of non-beneficial
452 cases excluded; 80% falsely flagged), reflecting the asymmetric cost structure:
453 missing a beneficial recalibration costs more than an unnecessary one. *Caveat*: the
454 threshold is searched on the same 1230-cell pool used for reporting (in-sample).
455 A leave-shift-type-out recomputation (threshold selected excluding the held-out
456 dose level, evaluated on it) yields $J = +0.17$ (downsampling), $J = -0.03$ (lead
457 drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127
458 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the
459 four held-out dose levels (results/deployment_holdout_recompute.csv).
460 The in-sample numbers are therefore optimistic, and the criterion requires inde-
461 pendent validation before operational use.

462 Every lookup-table input is *audit-time* (raw ECE, ID–OOD accuracy gap,
463 per-cell labels), requiring target ground-truth labels unavailable in the unlabeled
464 stream. The only label-free proxies examined are the predictability experiment’s
465 target statistics (predicted-prior L_1 distance, logit moments); that experiment trig-
466 gered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Ta-
467 ble ??), so no label-free deployable criterion can be claimed. The lookup table
468 is an *analysis- and audit-time decision aid* for a model owner with labeled eval-
469 uation data; the actionable deployment path is small-sample recalibration: buy n
470 labeled target records and recalibrate locally (protocol §5 S2). A deployable gate
471 remains future work (Limitations xiii).

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE > threshold → predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

472 Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfy-
473 ing the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 ,
474 point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion
475 was not evaluated), a safety rate of 0.0064. Sign convention: $\Delta\text{ECE} = \text{post-}$
476 $\text{recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the
477 primary-endpoint convention. *Coverage note:* the L2 shift matrix has been com-
478 pleted on all 390 planned (pair, seed, shift) combinations ($6 \text{ pairs} \times 5 \text{ seeds} \times 13$
479 levels ; results/l2_shift_full_390cells.csv); the safety figures quoted here are from
480 the initial two-seed InceptionTime subset (seeds 42–43, ResNet1D L2 excluded),
481 so these numbers rest on 2-seed, single-architecture coverage. The criterion is
482 rarely satisfied, so TS is a useful default but not a universally safe recalibrator,
483 and should be combined with local validation.

484 At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 ver-
485 sus 0.0999 for treating all cells; by the 90th percentile it turns negative
486 (-0.0024). Net benefit over treat-all is modest, confined to low thresholds
487 (results/deployment_metrics.csv).

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	−0.0145	0.0099	157
always-buy- n -labels	−0.2905	0.1861	1230
lookup-table	−0.0160	0.0289	155

488 Trivial strategy comparison (Table ??): the lookup-table strategy (mean ΔECE
489 −0.0160) outperforms always-none (0.000) and matches always-TS (−0.0145)
490 on the beneficial subset, but is dominated by always-buy- n -labels (−0.2905), the
491 target-label upper bound. The lookup table needs no target labels and captures
492 most of the TS benefit.

493 5.7. Noise floor control: SmoothECE differential under perfect calibration

494 A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an arti-
495 fact of SmoothECE’s sensitivity to temperature scaling T under perfect calibra-
496 tion ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated
497 predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute $\Delta\text{ECE}_{\text{noise}} =$
498 $\text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uni-
499 form, normal-skewed, bimodal) at $n = 2,000$.

500 The noise floor grows with T and is distribution-dependent: $T = 1.0$ gives
501 $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ gives +0.014—+0.035, $T = 1.5$

gives $+0.030$ – $+0.061$, $T = 3.0$ gives $+0.095$ – $+0.136$, and $T = 4.5$ gives $+0.126$ – $+0.164$. Observed raw ECE across the 62 raw-stage cells of the TS-component ablation (`results/ablation_ts_components.csv`, `stage=raw`) ranges 0.036 to 0.469 (mean 0.273), i.e. 2 – $28\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show large raw ECE (0.08–0.24), confirming the negative ΔECE reflects genuine TS-induced harm, not metric noise. For large $T > 3$ (noise floor 0.10–0.16) the floor is comparable to the observed $|\Delta\text{ECE}|$ in some seeds, so a metric-noise contribution cannot be excluded there; however, its *direction* (TS increases ECE under perfect calibration) is opposite to the positive $\Delta\text{ECE}_{\text{OOD}}$ in 51/60 experiments, so it cannot explain the positive findings. The primary endpoint is therefore a real effect, not a SmoothECE artifact, with the caveat that for large- T experiments the noise floor is non-negligible.

5.8. Discrimination-aware calibration gate (C5)

Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for deployment: a model whose discrimination has collapsed under shift may be recalibrable but clinically useless. We define a two-layer gate:

1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-class accuracy.

Across 12 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures), 7/12 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC, CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (InceptionTime only). The other 5/12 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls

below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$): TS improves calibration but discrimination has degraded too far to be clinically actionable.

This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes, Layer 2 rejects). It supplements C1 by also asking whether discrimination is sufficient for the recalibrated predictions to be clinically useful.

5.9. Real-data Shapley cross-fitting and predictability boundary

The synthetic Shapley validation (C3) confirms decomposition recovery in the identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We cross-fit all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

The linear model predicts the *direction* correctly in 51/60 (85%) but systematically *overestimates* the magnitude (mean prediction error +0.29, median +0.29). This is consistent with a predictability boundary: the decomposition captures the qualitative direction of the TS benefit but not its quantitative magnitude under real shift. The 9/60 direction errors cluster in CPSC $\rightarrow$ Chapman and Chapman $\rightarrow$ PTB-XL, where the Wasserstein-1 distance between source and target logit distributions is largest (mean 0.75 vs. 0.53 for correctly predicted directions), suggesting the linear approximation breaks down once the shift invalidates its underlying local Taylor expansion.

This strengthens the predictability failure branch (C3/C4): the Shapley decomposition is a valid *diagnostic* (*whether* TS will help) but not a valid *predic-*

551 *tor (how much)*. For clinical translation, the gate in §??, using only the sign of
552 ΔECE_{OOD} and the discrimination criterion, is the appropriate criterion, not the
553 Shapley-predicted magnitude.

554 5.10. Net benefit decision curve summary

555 Across all 60 seed experiments, the net benefit gain of TS over raw predictions
556 (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60
557 (83.3%) of experiments, with mean gain $+0.0027$ and range $[-0.001, +0.017]$.
558 The directions with the largest decision-cost reduction are CPSC→PTB-XL
559 (mean DCR $+0.0625$; net clinical value $+0.0312$) and CPSC→Chapman (mean
560 DCR $+0.0375$; NCV $+0.0188$), which also have the largest E3-subsample
561 ΔECE_{OOD} (mean $+0.137$ and $+0.136$ respectively, under the E3 subsampled
562 binned-ECE protocol with $n_{\text{id}} = n_{\text{ood}} = 20$ patients). The direction with negative
563 net benefit is Chapman→CPSC (mean DCR -0.0150 ; NCV -0.0075), which
564 is correctly rejected by the discrimination gate (C5). We note that the decision-
565 benefit ranking differs from the main-endpoint ΔECE_{OOD} ranking (which peaks
566 at PTB-XL→Chapman and PTB-XL→CPSC, means $+0.0240$ and $+0.0238$ on
567 the full SmoothECE grid): the two protocols differ in estimator, class set, and
568 subsample, so we treat the E3 analysis as a directionally supportive but protocol-
569 distinct check, not a conversion of the primary endpoint. What survives is
570 the gate-level conclusion: where the discrimination gate permits deployment,
571 the calibration benefit does not cost decision performance; where it is negative
572 (Chapman→CPSC), no calibration benefit should be deployed at all.

573 *5.11. Net Clinical Value (NCV) results*

574 The pre-registered NCV (Sec. ??) is BH-FDR corrected ($q = 0.05$) across the
575 two secondary endpoints (DCR, NCV).

576 Result: the OOD NCV is $+0.008125$ with 95% CI $[+0.0014, +0.0148]$
577 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS
578 yields a net improvement in OOD clinical decisions. The ID NCV is -0.006292
579 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *negative*: TS slightly
580 worsens ID decisions, consistent with the ID boundary condition (TS compresses
581 confidence toward the base rate). OOD NCV Cohen’s $d = +0.31$ and ID NCV
582 Cohen’s $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring
583 the main endpoint. Per-experiment NCV ranges over the 60 seeds are in the E3
584 CSV (`c1_dcr_ncv_multiclass.csv`).

585 *5.12. Reliability Diagrams*

586 Figure ?? shows the 6-direction averaged reliability curve, comparing model
587 confidence against empirical accuracy before and after temperature scaling. The
588 summary curve uses bin-count weighted averaging across all 6 transfer directions.

589 Figures ??–?? show the per-direction reliability diagrams for the representa-
590 tive configuration (ResNet1D, seed 42). In each panel, the red squares show the
591 raw model calibration curve and the green circles show the post-TS calibration
592 curve. The diagonal dashed line indicates perfect calibration. Grey bars (right
593 axis) show the sample count per bin. Error bars are bootstrap 95% percentile CIs
594 (1000 resamples).

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

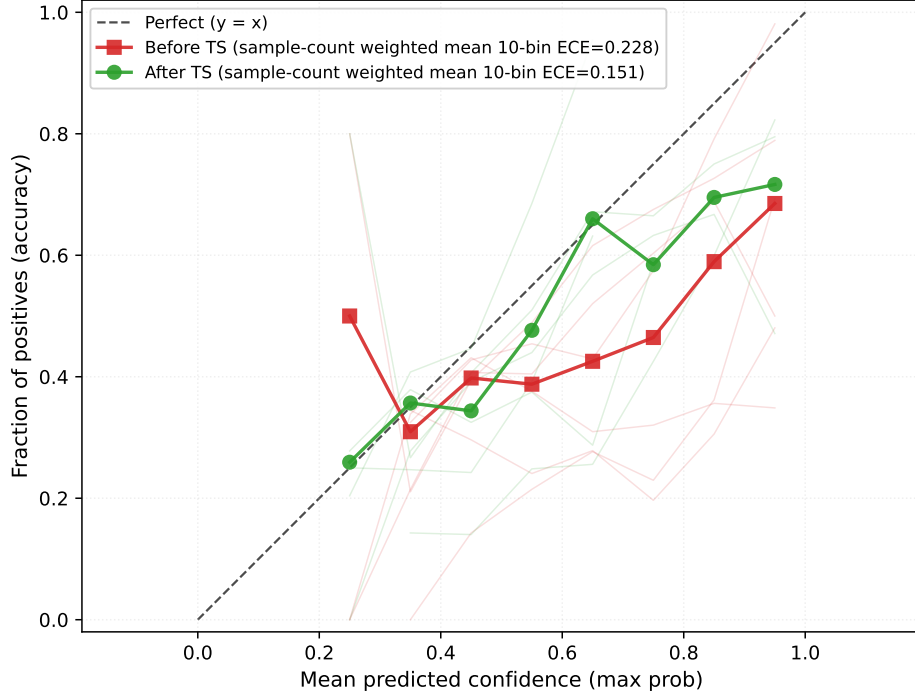


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

595 6. Discussion

596 6.1. OOD benefit seed variation and the nine counter-examples

597 The 51/60 robustness rate is driven by nine counter-examples across 5 of the
 598 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL
 599 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-
 600 XL→CPSC has zero). We transparently disclose all nine rather than excluding

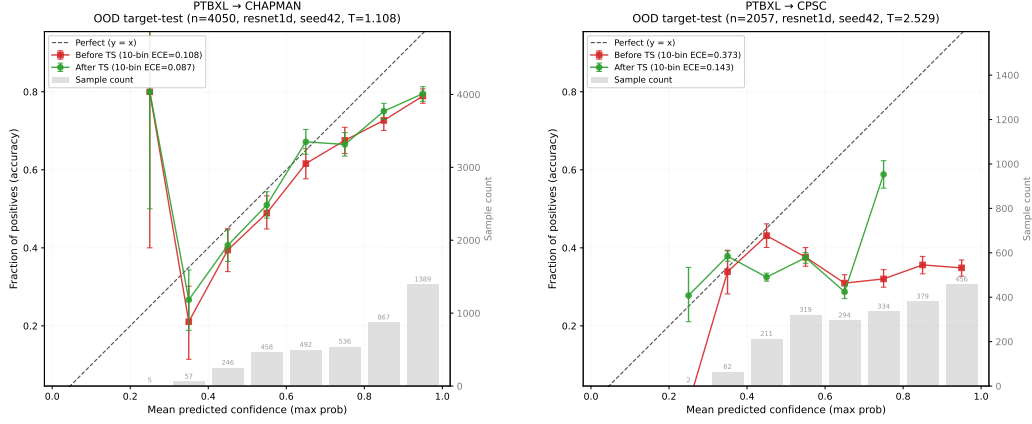


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

601 them, because the protocol requires all seed results to be reported and each has an
 602 identifiable post-hoc mechanism supporting the qualified claim.

603 Counter-example 1: Chapman→CPSC / InceptionTime / seed 45.
 604 $\Delta ECE_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur:
 605 lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44,
 606 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature
 607 mismatch, not room shortage); and negative decay (-0.0075), the source-fit
 608 temperature amplifying rather than correcting residual miscalibration.

609 Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} =$
 610 -0.0006 (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.001$) and at-
 611 tributable to finite-sample variation in the calibration split. The OOD raw ECE
 612 is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*); the mechanism is
 613 seed-level variation, not room shortage.

614 Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42.
 615 $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy is the low-

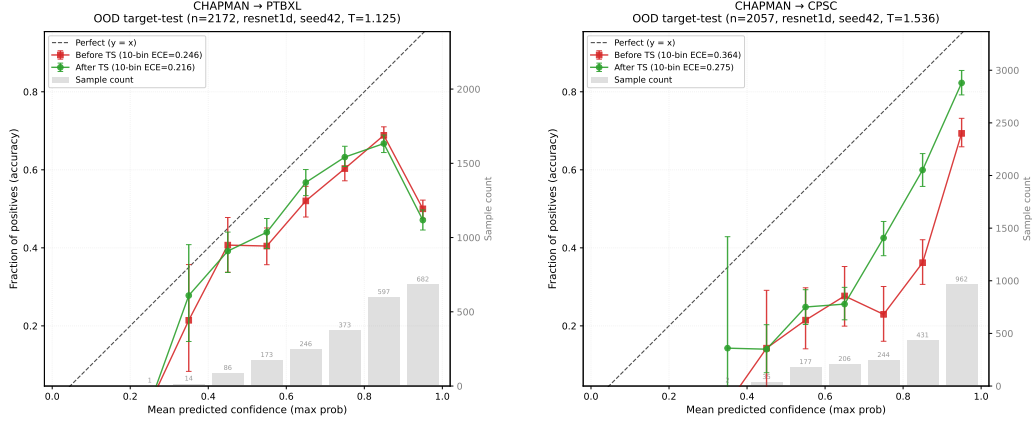


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

est across the 5 seeds (0.4655) and $\Delta\text{ECE}_{\text{ID}} = +0.0004$ is near zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL target.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. $\Delta\text{ECE}_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and $\Delta\text{ECE}_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already well-calibrated on ID, leaving nothing to correct.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42. $\Delta\text{ECE}_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta\text{ECE}_{\text{OOD}}| < 0.002$); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE (0.0649), and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. $\Delta\text{ECE}_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta\text{ECE}_{\text{ID}} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5

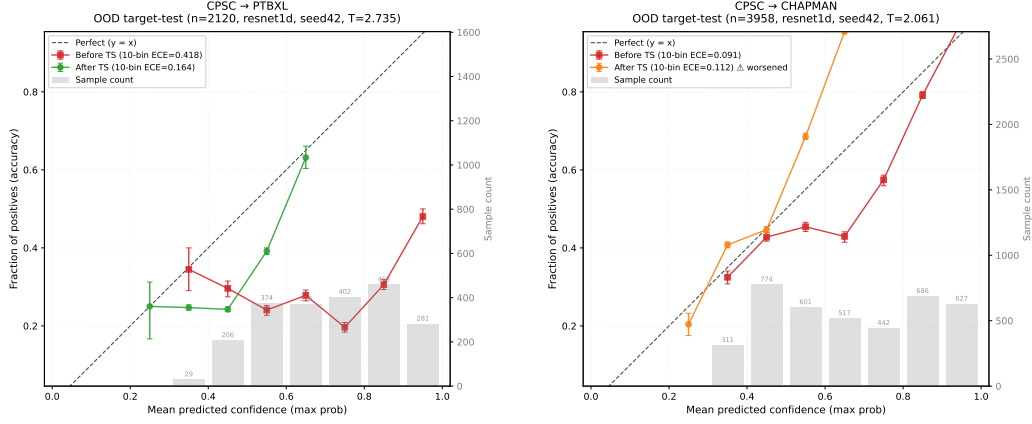


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-calibrated distribution degrades both ID and OOD calibration), unlike the “low OOD accuracy + large gap” mechanism of the other 8.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. $\Delta ECE_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and $\Delta ECE_{\text{ID}} = -0.0032$ is slightly negative, leaving nothing to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. $\Delta ECE_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983 (lowest across the 5 seeds) and $\Delta ECE_{\text{ID}} = -0.0011$ is slightly negative, again leaving nothing to correct.

Common pattern: eight of nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temper-

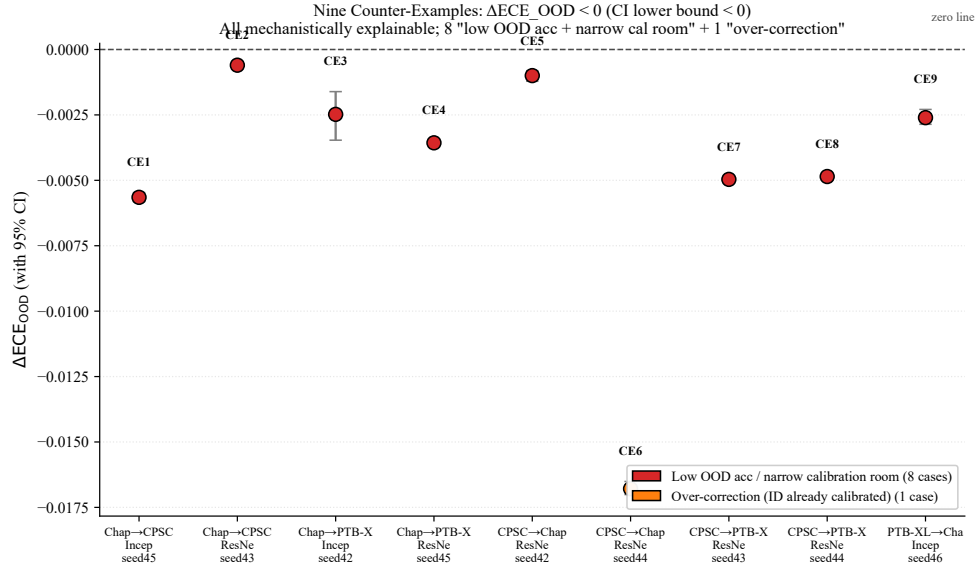


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

ature mismatch; the ninth (CE6) is over-correction on an already well-calibrated backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so the failure mechanism is seed-level temperature mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variation; the other 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

The 15.0% rate exceeds the 5% threshold common in clinical safety deployment. As a methodological boundary investigation, not a clinical deployment guide, the relevant criterion is whether the effect is detectable and the mechanism explainable, not a zero failure rate. All 9 counter-examples have identifiable mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”), and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

6.2. *Prior correction vs. posterior correction robustness*

The method-selection boundary (Table ??) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30 and 24/30 support respectively with small variance; prior-correction methods achieve higher mean ΔECE_{OOD} on the PTB-XL→CPSC pair (EM +0.0449, BBSE +0.0777) but much higher variance on the Chapman→CPSC pair (EM CI lower bound minimum −0.0763, BBSE −0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

670 6.3. *Boundary condition: the ID benefit is small but nonzero*

671 The ID boundary condition (C2) has a direct interpretation under the triggered
672 branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing zero; the observed
673 rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a small but significant
674 positive ID benefit (mean $+0.0083$ vs. OOD $+0.0159$; stratum means -0.0008 to
675 $+0.0232$, Table ??). The boundary condition does *not* mean TS is ineffective on
676 ID nor that the ID benefit is zero: the ID benefit is on average about half the OOD
677 benefit, so part of the apparent OOD gain is shared with the ID domain. A val-
678 ECE early-stopping effect (model selection within the endpoint family) plausibly
679 contributes to the positive ID estimates, opposite to the protocol’s val-NLL rule (a
680 registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores
681 the decay contrast $= \Delta \text{ECE}_{\text{OOD}} - \Delta \text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is
682 qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively
683 OOD effect. The ID-vs-OD asymmetry is therefore a *difference in magnitude*
684 between two positive benefits, not presence versus absence.

685 6.4. *Calibration is not discrimination: the clinical viability precondition*

686 Temperature scaling is a monotone transform of the softmax probabilities;
687 it preserves the argmax by construction (asserted bitwise in the released sanity
688 check), so it cannot change classification accuracy, AUROC, or AUPRC. The pri-
689 mary endpoint $\Delta \text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes
690 an already-deployable model *trustworthy*; it cannot make a non-discriminative
691 model deployable.

692 The accuracy statistics make this concrete. Across the 60 TS experiments,
693 mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean

ID-ODD drop is 0.236 (maximum 0.412). Against each target corpus’s majority-
 class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), mean
 OOD accuracy is *below* baseline in 3 of 6 directions, with a fourth essentially
 at baseline: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411
 (0.013 above baseline); CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620
 (both well below 0.713). In those directions recalibration is moot, since no tem-
 perature improves a model that should be replaced by a majority-class predictor,
 consistent with the counter-example pattern (6/9 with low OOD accuracy). The re-
 leased artifacts include per-direction AUROC/AUPRC for all 60 experiments (re-
 sults/discrimination_metrics_60exp.csv): mean OOD AUROC ranges from 0.629
 (CPSC→Chapman) to 0.795 (PTB-XL→Chapman), i.e. far above the accuracy-
 based majority-class reading yet direction-dependent; and TS leaves AUROC un-
 changed (mean $\Delta\text{AUROC}_{\text{TS-raw}} = -0.001$, max |change| 0.024), as the monotone
 transform requires. The discrimination-first gate—measure target discrimination
 before investing in recalibration—precedes the calibration decision rule in deploy-
 ment.

6.5. *Explainable $\neq$ predictable: post-hoc vs. pre-registered*

We distinguish *explainable* from *predictable*:

- *Explainable* (post-hoc, mechanism-level): after observing a counter-
 example, we identify the co-occurring conditions (low OOD accuracy, low
 OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. ??),
 a post-hoc explanation, not a pre-registered prediction.
- *Predictable* (pre-registered, deployment-time): a target-domain unsuper-
 vised statistic (predicted-prior L_1 distance, logit moments) predicts the ben-

718 efit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 .
719 The predictability experiment (Sec. ??) triggers the failure branch (LOO R^2
720 CI upper bound $0.104 < 0.5$).

721 The counter-examples are explainable but not predictable: the unsupervised target-
722 domain statistics available at deployment time cannot flag them in advance, which
723 is why the predictability experiment triggers the failure branch despite mechanis-
724 tic explainability. The two-step chain (decomposition + deployment lookup table)
725 provides post-hoc explanation and actionable guidance without a pre-registered
726 predictability claim.

727 6.6. *Transparent re-location vs. HARKing*

728 The pre-registration revision A1 (primary endpoint re-location from decay to
729 $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [?, ?] is
730 whether the revision direction is theory-driven or result-driven. Our revision di-
731 rection (“OOD is the calibration benefit source”) is consistent with the empirically
732 observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we
733 explicitly frame as an empirical prior rather than a theorem; the exploratory pilot
734 result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The
735 pilot results do not enter the main test report (protocol §7). All hypotheses in this
736 paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect
737 the post-revision status.

738 6.7. *Net clinical value: implications of the directional NCV asymmetry*

739 The NCV (Sec. ??) shows a directional asymmetry: OOD NCV = $+0.008125$
740 ($p = 0.019$) is a small but significant positive value (recalibration improves de-
741 cisions on unseen target data more often than it degrades them); ID NCV =

742 -0.006292 ($p = 0.016$) is a small but significant negative value (a minor decision-
743 cost penalty on data the source model already calibrates well).

744 Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60
745 positive) is supported: the positive OOD NCV shows the calibration gain trans-
746 lates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost**
747 **improvement claim** (C5: 7/12 strata deployable) is partially qualified: the nega-
748 tive ID NCV tempers source-domain deployability. We therefore frame the NCV
749 as *supportive of OOD deployment* and *cautionary for ID deployment*. It also in-
750 forms the meta-analytic pooling (Sec. ??), where the asymmetry contributes to
751 CI-width heterogeneity across splits. A permutation test with tighter CIs and a
752 multi-threshold NCV analysis are registered as future work.

753 6.8. Limitations and future work

754 Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on
755 all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); Bi-
756 Mamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on an
757 RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to
758 two-step); the deployment criterion is an empirical lookup table, not a predictive
759 criterion; (iii) real-data Shapley cross-fitting (§??) shows 85% direction accuracy
760 but systematic magnitude overestimation (mean error $+0.29$), and a sensitivity
761 analysis with T , Wasserstein-1, and entropy features confirms the predictability
762 boundary is robust (all $R^2 < 0$); (iv) the 27-cell decomposition validation uses
763 binned ECE, not the main endpoint’s SmoothECE, so quantitative conclusions
764 cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely
765 satisfied on the full L2 shift matrix (TS safety rate 0.0064); TS is a useful de-
766 fault, not universally safe; (vi) nine counter-examples of 60 experiments (15.0%)

767 are transparently disclosed, with a frequent post-hoc pattern (low OOD accuracy,
 768 non-positive decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the
 769 operating CI is BCa at $B = 10,000$, re-run on all 60 experiments (60/60 success-
 770 ful, metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 ex-
 771 periments are retained as historical provenance and are no longer used for any
 772 reported number. All 60 main-grid cells carry complete $B = 10,000$ /BCa prove-
 773 nance; the only cells lacking such metadata are the two excluded BiMamba toy
 774 runs (PTB-XL→Chapman, seeds 42–43), which do not enter the main endpoint.
 775 (viii) the Youden threshold is fitted in-sample (the protocol’s held-out dose lev-
 776 els were not used), making the operating characteristic optimistic; (ix) the L2
 777 deployment/safety grid has since been completed on all 390 planned (pair, seed,
 778 shift) cells (`results/l2_shift_full_390cells.csv`), and the L2 noise levels are syn-
 779 thetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavail-
 780 able under PhysioNet licensing); (x) temperature was fitted once per experiment,
 781 so CIs omit temperature-fitting uncertainty (the pre-registered two-layer bootstrap
 782 is implemented but not enabled); (xi) model selection used validation ECE rather
 783 than the pre-registered NLL early-stopping rule, plausibly contributing to the pos-
 784 itive ID boundary estimates; (xii) the per-pair decomposition into covariate, prior-
 785 probability, and concept-shift components is not asserted, since each pair mixes all
 786 three, isolated at eval time; (xiii) the deployment lookup table requires labeled tar-
 787 get data, and the label-free proxies failed their pre-registered branch, so no label-
 788 free gate is claimed; (xiv) TS is argmax-invariant and cannot improve accuracy,
 789 AUROC, or AUPRC; the artifacts now include per-direction AUROC/AUPRC
 790 for all 60 experiments (`results/discrimination_metrics_60exp.csv`), so the viabil-
 791 ity precondition is screened by both accuracy and discrimination (OOD AUROC

means range 0.629–0.792 by direction), with OOD accuracy below the majority-
 class baseline in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is direc-
 tionally asymmetric (OOD = +0.008125, $p = 0.019$, positive; ID = −0.006292,
 $p = 0.016$, negative), qualifying the C5 claim for source-domain but supporting
 target-domain deployment; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$
 to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits,
 with 48 as future work; (xvii) this work is from a single-author study, limiting inde-
 pendent review and audit, partially mitigated by the pre-registration protocol and
 public repository; (xviii) the secondary endpoint G (recoverable loss versus an or-
 acle recalibrator) is *not delivered*: `fit_oracle/apply_oracle` are implemented
 but were never executed on real data, so G is undefined in this study and no G
 result is reported anywhere in this paper or the repository; (xix) no signal-quality-
 index (SQI) stratification is performed: the target corpora are pipeline-processed
 releases and the L2 noise levels are synthetic, so the calibration benefit is not
 characterised as a function of measured signal quality (pSQI/kSQI-type indices);
 this is the principal gap for a signal-processing readership and is the first item of
 future work; (xx) no test–retest or repeatability analysis of the fitted temperature
 is possible within this data regime: all three public corpora are cross-sectional
 and provide a single recording per patient (Chapman-Shaoxing reports exactly
 one ECG per patient, §??), so same-patient across-session stability of T cannot
 be estimated from these data and is deferred to prospective multi-session collec-
 tion. Future work: (1) prospective multi-device validation; (2) continuous shift
 spaces; (3) π -independent recovery estimators (BBSE/EM-class) to upgrade the
 MAPE_{π} to a real estimator test; (4) the ResNet1D deficit on CPSC→Chapman and
 CPSC→PTB-XL (3/5 each) to test for a systematic interaction; (5) confirmatory

817 testing under the A2 reduction (12 hypotheses, BH-12 $q = 0.05$, 9/12 significant,
818 Bonferroni-12 = 2/12); (6) re-running the endpoint with $n_{\text{filters}} = 48$ Inception-
819 Time on a higher-memory GPU; (7) a multi-threshold permutation test on NCV.

820 7. Conclusion

821 We pre-registered and validated a multi-seed calibration boundary study of
822 temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD benefit: 51/60
823 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime 27/30 = 90.0%,
824 ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID boundary: 23/60 cells
825 (38.3%) have 95% CIs containing zero under the BCa operating CI (percentile:
826 29/60 = 48.3%), below the $\geq 80\%$ criterion, triggering the pre-registered branch;
827 the OOD benefit is $1.9\times$ the ID benefit. **(C3)** Shapley decomposition: 27/27 cells
828 pass at $n = 20,000$ (21/27 at $n = 500$). **(C4)** Method selection: TS robust (51/60),
829 EM prior and Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/12 strata pass both
830 layers, reframing the 9 counter-examples as 4 calibration and 5 discrimination
831 failures. The predictability failure branch is triggered (three-step chain becomes
832 two-step), and all hypotheses are exploratory + robustness validation, not confir-
833 matory; protocol v2.1-A1 applies. TS needs local validation (safety rate 0.0064),
834 and OOD accuracy is below the majority-class baseline in 3 of 6 directions.

835 A noise floor control (§??) bounds, but does not eliminate, a SmoothECE ar-
836 tifact: at moderate temperatures metric noise is $2\text{--}28\times$ below the observed raw
837 ECE (0.036–0.469, mean 0.273), whereas for large $T > 3$ the floor (0.10–0.16)
838 is comparable to the observed $|\Delta\text{ECE}|$ and a metric-noise contribution cannot be
839 excluded there. Real-data Shapley cross-fitting (§??) gives 85% direction accu-

840 racy with magnitude overestimation (mean error +0.29); T , Wasserstein-1, and
841 entropy features confirm the boundary (all $R^2 < 0$).

842 The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative
843 (-0.000573) because narrow-CI studies dominate, while DerSimonian–Laird ran-
844 dom effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossAr-
845 chitecture +0.0148). We caution, however, that the meta-analytic independence
846 assumption is *not* satisfied here: the 60 cells share the same three corpora,
847 patient sets, and architectures, and their between-cell Q statistics are extreme
848 ($I^2 \approx 99.98\%$), so both FE and DL confidence intervals are spuriously narrow.
849 The interpretable pooled uncertainty is therefore the cluster-robust interval over
850 the 12 pair $\times$ architecture strata: $\Delta\text{ECE}_{\text{OOD}} = +0.0159$ [0.0113, 0.0205] (t, df= 11;
851 stratum SD 0.0072, about $1800\times$ the single-experiment bootstrap SE). The point
852 estimate 0.0148–0.0159 and the count-based proportion ($51/60 = 85.0\%$) stand.

853 The 60 experiments form the robustness-evidence family; amendment A2 re-
854 duces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypothe-
855 ses ($6 \text{ pairs} \times 2 \text{ architectures} \times 1 \text{ primary shift level}$), with all 13 L2 shift levels
856 re-designated as exploratory dose-response. BH-60 and Bonferroni leave the ro-
857 bustness headline unchanged ($51/60$); the confirmatory family is BH-12 with 9/12
858 significant, and Bonferroni-12 retains 2/12. Across the 60 TS experiments the
859 95% CI widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells
860 fall below the narrow-CI diagnostic threshold (0.005) and six supporting cells
861 have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60
862 (75.0%). Narrow CIs reflect the moderate calibration sample and low SmoothECE
863 variance; they do not affect the sign-based support/counter-example classification.

864 **Honesty Declaration (5 items, adversarial audit product)**

- 865 1. MAPE_π is a **design check** (resampling constant read-back), not an estimator capability test; independent π recovery (BBSE/EM) is a deferred item
866 (protocol §13).
867
- 868 2. Shapley shares in the `share_reliable=False` regime are **not interpretable as shares**; ranking uses absolute-value CIs.
869
- 870 3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the
871 archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information
872 mode), reported via exit code 2.
- 873 4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
874 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement
875 demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13).
876 The wording is limited to “mechanism decomposition (synthetic validation
877 part)”.
- 878 5. The repository’s `scripts/preprocess_cinc2021.py`,
879 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
880 from the third-party SafeECGMatch release (original README in
881 `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment
882 grid.

883 **Data availability**

884 The PTB-XL [?], Chapman-Shaoxing [?], and CPSC2018+2019
885 datasets are publicly available via PhysioNet/Kaggle under their respective
886 licenses. The pre-registered protocol v2.1-A1, its hash manifest
887 (`docs/osf_archive_manifest.json`), the revision A1 document,

888 and the full per-experiment result files (checkpoints/transfer/**/
889 transfer_result.json, results/*.csv) are included with the submis-
890 sion as supplementary material and are also available in the public repository
891 <https://github.com/Toni-cs/ecg-calibration-transportability>.
892 The protocol, both amendments (A1, A2), and a per-file archival-status re-
893 port are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
894 docs/osf_archive_manifest.json is the internal snapshot taken on 2026-09-
895 05; it is not a substitute for the public record.

896 **Code availability**

897 The full pipeline code, preprocessing (reproducible from the public source
898 databases), patient-level split generation, model training, the calibration-method
899 implementations, and all statistical analyses, together with the environment
900 lock file, the patient-level split indices for all five dataset variants and five
901 random seeds, and selected per-experiment result artifacts, is publicly available
902 at <https://github.com/Toni-cs/ecg-calibration-transportability>.
903 Trained model checkpoints for the 60-experiment transfer grid are provided as
904 release assets of the same repository. The repository will be archived on Zenodo
905 with a DOI upon acceptance. No new raw waveforms were collected or generated;
906 the patient-level split indices and label-mapping tables released with the code
907 fully determine the data configuration used in every experiment.

908 **Declaration of generative AI and AI-assisted technologies in the writing pro-**
909 **cess**

910 All scientific work reported in this study—the study design, the pre-registered
911 protocol and its amendments, the implementation and execution of the 60-
912 experiment transfer grid, the statistical analysis, the interpretation of the results,
913 and the writing of the manuscript—was performed by the author. During the final
914 preparation of the manuscript, the author used a large language model assistant
915 (GLM) solely as a language-editing aid, to improve wording and readability of
916 text the author had written. It was not used to generate scientific content, exper-
917 imental data, analyses, figures, or conclusions. After using this tool, the author
918 reviewed and edited the text as needed and takes full responsibility for the content
919 of the published article.

920 **CRediT authorship contribution statement**

921 **Toni Guan:** Conceptualization, Methodology, Software, Validation, Formal
922 analysis, Investigation, Data curation, Writing – original draft, Writing – review
923 & editing, Visualization, Project administration.

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931 **Declaration of competing interest**

932 The author declares no competing financial or personal interests.

933 The 60 per-seed reliability diagrams (6 transfer pairs $\times$ 2 architectures $\times$ 5
934 seeds; calibration curves, reliability diagrams, and Δ ECE bars per experiment)
935 are provided as a separate supplementary file (supplementary .pdf, §1) and are
936 additionally released with the code repository.

937 **References**

- 938 [1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon,
939 B. Lakshminarayanan, J. Snoek, “Can you trust your model’s uncertainty?
940 Evaluating predictive uncertainty under dataset shift,” *NeurIPS*, 2019.
941 arXiv:1906.02530.
- 942 [2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, “On calibration of modern neural
943 networks,” *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.
- 944 [3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., “PTB-XL, a large
945 publicly available electrocardiography dataset,” *Scientific Data*, 7:154,
946 2020. doi:10.1038/s41597-020-0495-6.
- 947 [4] J. Zheng, J. Guo, H. Chu, “A large scale 12-lead electrocardiogram
948 database for arrhythmia study (Chapman-Shaoxing e-cardiology),” version
949 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.

- 950 [5] A. L. Goldberger et al., “PhysioBank, PhysioToolkit, and PhysioNet:
951 components of a new research resource for complex physiologic signals,”
952 *Circulation*, 101(23):e215–e220, 2000.
- 953 [6] J. Nixon, M. W. Dusenberry, L. Zhang, et al., “Measuring calibration in
954 deep learning,” *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.
- 955 [7] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach,
956 “Beyond temperature scaling: obtaining well-calibrated multi-class
957 probabilities with Dirichlet calibration,” *NeurIPS*, 2019. arXiv:1910.12656.
- 958 [8] A. Alexandari, A. Kundaje, A. Shrikumar, “Maximum likelihood with
959 bias-corrected calibration is hard-to-beat at label shift adaptation,” *ICML*,
960 PMLR 119:222–232, 2020. arXiv:1901.06852.
- 961 [9] Z. Lipton, Y.-X. Wang, A. Smola, “Detecting and correcting for label shift
962 with black box predictors,” *ICML*, PMLR 80:1944–1953, 2018.
963 arXiv:1802.03916.
- 964 [10] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão,
965 F. Cabitza, H. Gamboa, “Evaluation of uncertainty quantification methods
966 in multi-label classification: a case study with automatic diagnosis of
967 electrocardiogram,” *Information Fusion*, 101:101978, 2024.
968 doi:10.1016/j.inffus.2023.101978.
- 969 [11] J. F. Vranken et al., “Uncertainty estimation for deep learning-based
970 automated analysis of 12-lead electrocardiograms,” *European Heart*
971 *Journal – Digital Health*, 2(3):423–434, 2021. doi:10.1093/ehjdh/ztab045.

- 972 [12] M. Reyna, N. Strodthoff, P. Wagner, et al., “Issues in the automated
973 classification of multilead ECGs using heterogeneous labels and
974 populations,” *Physiological Measurement*, 43(8):084001, 2022.
975 doi:10.1088/1361-6579/ac79fd.
- 976 [13] B. Efron, “Better bootstrap confidence intervals,” *Journal of the American*
977 *Statistical Association*, 82(397):171–185, 1987.
978 doi:10.1080/01621459.1987.10478410.
- 979 [14] B. Van Calster, D. J. Steyerberg, S. Collins, R. D. Riley, E. W. Steyer, “A
980 calibration hierarchy for risk models was defined: from utopia to empirical
981 data,” *Journal of Clinical Epidemiology*, 74:167–176, 2016.
982 doi:10.1016/j.jclinepi.2015.12.005.
- 983 [15] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, G. S. Collins,
984 “Calibration: the Achilles heel of predictive analytics,” *BMC Medicine*,
985 17:230, 2019. doi:10.1186/s12916-019-1466-7.
- 986 [16] A. Kumar, P. Liang, T. Ma, “Verified uncertainty calibration,” *NeurIPS*,
987 2019. arXiv:1909.10155.
- 988 [17] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, “Deep learning for ECG
989 analysis: benchmarks and insights from PTB-XL,” *IEEE Journal of*
990 *Biomedical and Health Informatics*, 25(5):1519–1528, 2021.
991 doi:10.1109/JBHI.2020.3022989.
- 992 [18] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto,
993 “RR-interval-based atrial fibrillation detection and burden estimation:

- 994 cross-dataset validation and calibration-aware probability analysis,”
 995 *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.
- 996 [19] K. Patel, P. Beedala, “Calibration drift under cross-institutional deployment:
 997 an external validation framework for ICU mortality prediction across
 998 MIMIC-IV and eICU,” *medRxiv* (preprint), 2026.
 999 doi:10.64898/2026.05.03.26352335.
- 1000 [20] X. Wang, M. Long, J. Wang, M. I. Jordan, “TransCal: transferability
 1001 estimation for auto-selecting transfer calibration,” *NeurIPS*, 2020.
 1002 arXiv:2007.08259.
- 1003 [21] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, “A unified view of label shift
 1004 estimation,” *NeurIPS*, 2020. arXiv:2003.07554.
- 1005 [22] T. Popordanoska, G. Radevski, T. Tuytelaars, et al., “LaSCal: label-shift
 1006 calibration for uncertainty quantification,” *NeurIPS*, 2024.
- 1007 [23] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla, F. Herrera,
 1008 “A unifying view on dataset shift in classification,” *Pattern Recognition*,
 1009 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.
- 1010 [24] [Pre-registered protocol, publicly archived] “ECG calibration boundary
 1011 study – experiment protocol,” 2026. Hash manifest:
 1012 docs/osf_archive_manifest.json (local snapshot; archived at
 1013 <https://doi.org/10.17605/OSF.IO/53H64>); revision A1 full text in
 1014 docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md.

- 1015 [25] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, “The preregistration
1016 revolution,” *PNAS*, 115(11):2600–2606, 2018.
1017 doi:10.1073/pnas.1708274114.
- 1018 [26] D. Lakens, “The practical alternative to the p value is the correctly used p
1019 value,” *Perspectives on Psychological Science*, 16(3):639–648, 2021.
1020 doi:10.1177/1745691620958012.
- 1021 [27] J. Li, A. D. Aguirre, V. M. Junior, et al., “An Electrocardiogram Foundation
1022 Model Built on over 10 Million Recordings,” *NEJM AI*, 2(7):AIoa2401033,
1023 2025. doi:10.1056/AIoa2401033.
- 1024 [28] J. C. McHugh, B. Wang, et al., “ECG-FM: an open electrocardiogram
1025 foundation model,” *arXiv preprint* 2408.05178, 2024.
- 1026 [29] J. Jin, et al., “Reading your heart: learning ECG words and sentences via
1027 pre-training ECG language model (HeartLang),” *ICLR*, 2025.
1028 arXiv:2502.10707.
- 1029 [30] C. V. Nguyen, C. D. Do, “Transfer learning in ECG diagnosis: is it
1030 effective?” *PLOS ONE*, 20(5):e0316043, 2025.
1031 doi:10.1371/journal.pone.0316043.
- 1032 [31] S. Vavaroutas, et al., “ADAPTOOD: uncertainty-aware fine-tuning for
1033 out-of-distribution ECG time series models,” *arXiv preprint* 2606.04164,
1034 2026.
- 1035 [32] W. Jiang, et al., “Test-time adaptation for ECG classification via SQI-gated
1036 self-training and beat-rhythm consistency,” *arXiv preprint* 2608.23347,
1037 2026 (accepted at MICCAI 2026).

- 1038 [33] N. Nemati, “Comparative analysis of data augmentation for clinical ECG
1039 classification with STAR (Sinusoidal Time–Amplitude Resampling),” *arXiv*
1040 *preprint* 2510.24740, 2025.
- 1041 [34] X. Liu, et al., “PEACE: knowledge-guided cross-modal fusion for
1042 adult-to-pediatric ECG transfer via label-conditioned contrastive
1043 alignment,” *arXiv preprint* 2605.00647, 2026.
- 1044 [35] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo, “Foundation
1045 models for electrocardiograms,” *arXiv preprint* 2407.07110, 2024.